WORKED SOLUTION

STEP III 2017, Question 4

Pure • Geometric Mean of Functions

STEP III 2017, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q4

The key idea

The geometric mean is just an exponential of an ordinary average: $\ln F(y)$ is the mean value of $\ln f$ over $[0, y]$. So every property of F comes from a property of logarithms (change of base, $\ln(fg) = \ln f + \ln g$) combined with linearity of the integral. For the converse in (iv), the condition becomes an equation involving an integral, and differentiating it turns it into a simple differential equation.

Solution

Throughout, write

$$\ln F(y) = \frac{1}{y}\int_0^y \ln f(x)\,\mathrm{d}x \qquad (y > 0),$$

which is the definition with logarithms taken of both sides.

Part (i)

Since $a > 0$ and $a \neq 1$, $\log_a$ is defined and $\ln a \neq 0$, and the change-of-base rule gives $\log_a f(x) = \dfrac{\ln f(x)}{\ln a}$. Also $a^t = \mathrm{e}^{t\ln a}$ for any real t. Hence

$$\begin{aligned} a^{\frac{1}{y}\int_0^y \log_a f(x)\,\mathrm{d}x} &= \exp\left(\ln a \cdot \frac{1}{y}\int_0^y \frac{\ln f(x)}{\ln a}\,\mathrm{d}x\right) \\ &= \exp\left(\frac{1}{y}\int_0^y \ln f(x)\,\mathrm{d}x\right) = F(y), \end{aligned}$$

where we took the constant $\frac{1}{\ln a}$ outside the integral and cancelled it against $\ln a$. This is the required result. (The condition $a \neq 1$ is exactly what lets us divide by $\ln a$.)

Part (ii)

Since $f(x) > 0$ and $g(x) > 0$, we have $h(x) > 0$, so H is defined, and $\ln h(x) = \ln f(x) + \ln g(x)$. By linearity of the integral,

$$\ln H(y) = \frac{1}{y}\int_0^y (\ln f(x) + \ln g(x))\,\mathrm{d}x = \ln F(y) + \ln G(y) = \ln(F(y)G(y)).$$

As ln is one-to-one, $H(y) = F(y)G(y)$.

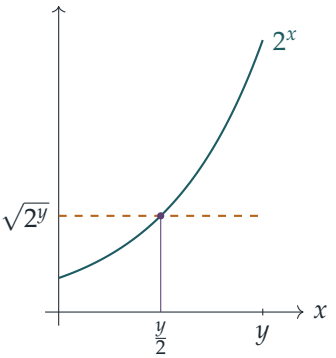
Part (iii)

Here $f(x) = b^x > 0$ and $\ln f(x) = x \ln b$, so

$$\ln F(y) = \frac{1}{y}\int_0^y x \ln b\,\mathrm{d}x = \frac{1}{y}\cdot\frac{y^2}{2}\ln b = \frac{y}{2}\ln b.$$

Therefore $F(y) = \mathrm{e}^{\frac{y}{2}\ln b} = b^{y/2} = \sqrt{b^y}$, as required.

The picture below shows what this says for $b = 2$: the geometric mean of 2^x over $[0, y]$ is the value of 2^x at the midpoint $\frac{y}{2}$, because $\ln 2^x$ is linear and the average of a linear function is its midpoint value.



Part (iv)

We assume, as the question implicitly does, that f is continuous, so that the integrals exist and we may use the fundamental theorem of calculus. Put $\phi(x) = \ln f(x)$, which is defined because $f(x) > 0$, and is continuous. The hypothesis $F(y) = \sqrt{f(y)}$ says, after taking logarithms, that

$$\frac{1}{y}\int_0^y \phi(x)\,\mathrm{d}x = \tfrac{1}{2}\phi(y) \quad \text{that is,} \quad \int_0^y \phi(x)\,\mathrm{d}x = \frac{y}{2}\,\phi(y) \qquad \text{for all } y > 0. \qquad (*)$$

Ask yourself How can we get rid of the integral, and are we allowed to?

Differentiate with respect to y. The left-hand side of $(*)$ is differentiable, with derivative $\phi(y)$, by the fundamental theorem of calculus. So the right-hand side is differentiable

too, and therefore so is $\phi(y) = \frac{2}{y} \times$ (left-hand side) for $y > 0$. Differentiating $(*)$ using the product rule on the right:

$$\phi(y) = \tfrac{1}{2}\phi(y) + \tfrac{y}{2}\phi'(y) \quad \Longrightarrow \quad y\,\phi'(y) - \phi(y) = 0.$$

Since $y > 0$ we may divide by y^2 and recognise a quotient-rule derivative:

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{\phi(y)}{y}\right) = \frac{y\phi'(y) - \phi(y)}{y^2} = 0.$$

So $\phi(y)/y$ is a constant, k say, on the interval $y > 0$ (a function with zero derivative on an interval is constant, by the mean value theorem). Hence $\ln f(x) = kx$ and

$$f(x) = \mathrm{e}^{kx} = b^x, \qquad \text{where } b = \mathrm{e}^k > 0.$$

This holds for every $x > 0$; the value at the single point $x = 0$ doesn't affect any of the integrals, and by continuity $f(0) = 1 = b^0$ anyway. Together with (iii), this shows that the functions whose geometric mean is $\sqrt{f(y)}$ are exactly the functions b^x.

Checks

- In (iii) with $b = 4$, $y = 2$: $\ln F(2) = \frac{1}{2}\int_0^2 x \ln 4 \,\mathrm{d}x = \ln 4$, so $F(2) = 4 = \sqrt{4^2}$; a numerical integration agrees.
- In (iv), try a function that is not of the form b^x, such as $f(x) = 3 \cdot 2^x$. Then $F(y) = 3 \cdot 2^{y/2}$ by (ii) and (iii), while $\sqrt{f(y)} = \sqrt{3} \cdot 2^{y/2}$, so the condition fails, as it must. Our argument would give $\phi(y)/y = \ln 2 + \frac{\ln 3}{y}$, which is not constant.
- Parts (i) and (ii) were confirmed by computer for random positive functions, values of y and bases a on both sides of 1.

Common slips

- In (iv), differentiating $(*)$ but forgetting the product rule on $\frac{y}{2}\phi(y)$, which loses the ϕ' term entirely.
- Concluding $f(x) = b^x$ from (iii) alone: (iii) shows that b^x has the property, not that nothing else does. Part (iv) is the converse and needs its own argument.
- In (i), not noting where $a \neq 1$ is needed: it is what makes $\log_a$ meaningful and $\ln a \neq 0$.

Beyond STEP

The continuous geometric mean is the integral version of $(x_1 x_2 \cdots x_n)^{1/n}$, and Jensen's inequality for the concave function ln shows that it never exceeds the ordinary mean $\frac{1}{y}\int_0^y f(x)\,\mathrm{d}x$, the continuous AM–GM inequality. Part (iv) is a small example of a

functional equation solved by turning it into a differential equation, which requires care about differentiability; in analysis courses you learn exactly when the fundamental theorem of calculus allows this. For the rigorous background, see Abbott, *Understanding Analysis*.

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